Merging African Neolithic datasets

2025-08-20

```
res_dir <- "../results/s00_merge_neo_dataset/"
```

This script merges the African Neolithic datasets from EGAD50000000006 to EGAD50000000011 into a single gen_tibble object.

EGAD50000000011

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_A_no_I_D.gt")) {
  NEO_E_TC_A <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_A_no_I_D.g
} else {
  NEO_E_TC_A <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_A.bed"
  nrow(NEO_E_TC_A) # 41
  count_loci(NEO_E_TC_A) # 2221421
  NEO_E_TC_A <- NEO_E_TC_A %>% select_loci(-all_of(c(which(show_loci(NEO_E_TC_A)$allele_ref %in% c("I",
  count_loci(NEO_E_TC_A)
  double_NA_NEO_E_TC_A <- which(is.na(show_loci(NEO_E_TC_A)$allele_ref) & is.na(show_loci(NEO_E_TC_A)$al
  NEO_E_TC_A <- NEO_E_TC_A %>% select_loci(-all_of(double_NA_NEO_E_TC_A))
  count_loci(NEO_E_TC_A)
  table(NEO_E_TC_A$population)
  dups <- NEO_E_TC_A %>% find_duplicated_loci()
  NEO_E_TC_A <- NEO_E_TC_A %>% select_loci(-all_of(dups))

  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_A_no_I_D.bk")) {
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_A_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_A_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_A_no_I_D.gt")
  }

  NEO_E_TC_A <- gt_update_backingfile(NEO_E_TC_A,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_A_no_I_D"
  )

  #gt_save(NEO_E_TC_A, file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_20040
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_B_no_I_D.gt")) {
  NEO_E_TC_B <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_B_no_I_D.g
```

```
} else {
  NEO_E_TC_B <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_B.bed"
  nrow(NEO_E_TC_B) # 59
  count_loci(NEO_E_TC_B) # 2221421
  NEO_E_TC_B <- NEO_E_TC_B %>% select_loci(-all_of(c(which(show_loci(NEO_E_TC_B)$allele_ref %in% c("I",
  count_loci(NEO_E_TC_B)
  double_NA_NEO_E_TC_B <- which(is.na(show_loci(NEO_E_TC_B)$allele_ref) & is.na(show_loci(NEO_E_TC_B)$a
  NEO_E_TC_B <- NEO_E_TC_B %>% select_loci(-all_of(double_NA_NEO_E_TC_B))
  count_loci(NEO_E_TC_B)
  table(NEO_E_TC_B$population)
  dups <- NEO_E_TC_B %>% find_duplicated_loci()
  NEO_E_TC_B <- NEO_E_TC_B %>% select_loci(-all_of(dups))

  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_B_no_I_D.bk")) {
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_B_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_B_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_B_no_I_D.gt")
  }
  NEO_E_TC_B <- gt_update_backingfile(NEO_E_TC_B,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_200401_B_no_I_D"
  )
  #gt_save(NEO_E_TC_B, file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_E__TC-2508_20040
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

Merge NEO_E

```
if (file.exists("../data/intermediate/rbind_data/NEO_E_TC_A_B.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_E_TC_A_B.gt")
  file.remove("../data/intermediate/rbind_data/NEO_E_TC_A_B.bk")
  file.remove("../data/intermediate/rbind_data/NEO_E_TC_A_B.rds")
}
```

```
## [1] TRUE
```

```
NEO_E <- rbind(NEO_E_TC_A, NEO_E_TC_B,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_E_TC_A_B"
)
```

```
rm(NEO_E_TC_A, NEO_E_TC_B, NEO_E)
gc()
```

```
##           used (Mb) gc trigger    (Mb)  max used    (Mb)
## Ncells 1217259 65.1    4539810  242.5   5674762  303.1
## Vcells 4185075 32.0  131707805 1004.9 164029920 1251.5
```

EGAD50000000006

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110_no_I_D.gt")) {
  NEO_B_SE <-
    gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110_no_I_D.gt")
} else {
  NEO_B_SE <-
    gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110.bed",
      allow_duplicates = TRUE, quiet = TRUE,
      valid_alleles = c("T", "G", "A", "C", "I", "D")
    )
  nrow(NEO_B_SE) # 155
  count_loci(NEO_B_SE) # 2221421
  NEO_B_SE <- NEO_B_SE %>% select_loci(-all_of(c(which(show_loci(NEO_B_SE)$allele_ref %in% c("I", "D"))
  count_loci(NEO_B_SE)
  double_NA_NEO_B_SE <- which(is.na(show_loci(NEO_B_SE)$allele_ref) & is.na(show_loci(NEO_B_SE)$allele_a
  NEO_B_SE <- NEO_B_SE %>% select_loci(-all_of(double_NA_NEO_B_SE))
  count_loci(NEO_B_SE)
  table(NEO_B_SE$population) # Zambia
  dups <- NEO_B_SE %>% find_duplicated_loci()
  NEO_B_SE <- NEO_B_SE %>% select_loci(-all_of(dups))


  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110_no_I_D.bk")) {
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110_no_I_D.gt")
  }

  NEO_B_SE <- gt_update_backingfile(NEO_B_SE,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110_no_I_D"
  )
  #gt_save(NEO_B_SE, file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__SE-2209_191110_
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_2019_no_I_D.gt"))
  NEO_B_TE <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_2019_no_I_D
} else {
  NEO_B_TE <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_2019.bed
  nrow(NEO_B_TE) # 1
  count_loci(NEO_B_TE) # 2225121
  NEO_B_TE <- NEO_B_TE %>% select_loci(-all_of(c(which(show_loci(NEO_B_TE)$allele_ref %in% c("I", "D"))
  count_loci(NEO_B_TE)
  double_NA_NEO_B_TE <- which(is.na(show_loci(NEO_B_TE)$allele_ref) & is.na(show_loci(NEO_B_TE)$allele_a
  NEO_B_TE <- NEO_B_TE %>% select_loci(-all_of(double_NA_NEO_B_TE))
  count_loci(NEO_B_TE)
  unique(NEO_B_TE$population) # "Zambia_Kgalagadi"
  dups <- NEO_B_TE %>% find_duplicated_loci()
  NEO_B_TE <- NEO_B_TE %>% select_loci(-all_of(dups))
```

```
  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_2019_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_2019_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_2019_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_2019_no_I_D.gt")
  }
  NEO_B_TE <- gt_update_backingfile(NEO_B_TE,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_2019_no_I_D"
  )
  #gt_save(NEO_B_TE, file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_B__TE-2567_201023_
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

Merge NEO_B

```
if (file.exists("../data/intermediate/rbind_data/NEO_B_SE_TE.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_B_SE_TE.gt")
  file.remove("../data/intermediate/rbind_data/NEO_B_SE_TE.bk")
  file.remove("../data/intermediate/rbind_data/NEO_B_SE_TE.rds")
}
```

```
## [1] TRUE
```

```
NEO_B <- rbind(NEO_B_SE, NEO_B_TE,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_B_SE_TE"
)
```

```
rm(NEO_B_SE, NEO_B_TE, NEO_B)
gc()
```

```
##           used (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells 1217349 65.1    4927072 263.2   6158839  329.0
## Vcells 4185754 32.0  105960769 808.5 164029920 1251.5
```

EGAD50000000007

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D.gt")) {
  NEO_F_OE <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D.gt")
} else {
  NEO_F_OE <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625.bed", al
  nrow(NEO_F_OE) # 29
  count_loci(NEO_F_OE) # 2280697
  NEO_F_OE <- NEO_F_OE %>% select_loci(-all_of(c(which(show_loci(NEO_F_OE)$allele_ref %in% c("I", "D"))
  count_loci(NEO_F_OE)
```

```
  double_NA_NEO_F_OE <- which(is.na(show_loci(NEO_F_OE)$allele_ref) & is.na(show_loci(NEO_F_OE)$allele_a
  NEO_F_OE <- NEO_F_OE %>% select_loci(-all_of(double_NA_NEO_F_OE))
  count_loci(NEO_F_OE)
  table(NEO_F_OE$population) # "Botswana_Ghanzi"
  dups <- NEO_F_OE %>% find_duplicated_loci()
  NEO_F_OE <- NEO_F_OE %>% select_loci(-all_of(dups))


  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D.bk")) {
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D.gt")
  }
  NEO_F_OE <- gt_update_backingfile(NEO_F_OE,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D"
  )
  #gt_save(NEO_F_OE,
  #  file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D"
  #)
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

EGAD50000000008

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112_no_I_D.gt")) {
  NEO_A_TI <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112_no_I_D.gt")
} else {
  NEO_A_TI <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112.bed", al
  nrow(NEO_A_TI) # 133
  count_loci(NEO_A_TI) # 2225121
  NEO_A_TI <- NEO_A_TI %>% select_loci(-all_of(c(which(show_loci(NEO_A_TI)$allele_ref %in% c("I", "D"))
  count_loci(NEO_A_TI)
  double_NA_NEO_A_TI <- which(is.na(show_loci(NEO_A_TI)$allele_ref) & is.na(show_loci(NEO_A_TI)$allele_a
  NEO_A_TI <- NEO_A_TI %>% select_loci(-all_of(double_NA_NEO_A_TI))
  count_loci(NEO_A_TI)
  table(NEO_A_TI$population) # "CAR_Banda"                  "SouthAfrica_Coloured"      "SouthAfrica_Xho
  # [4] "Namibia_Damara"            "Namibia_Himba"             "Angola_Vasekele"
  # [7] "Namibia_TsumkweKung"       "SouthAfrica_Khomani"       "SouthAfrica_SothoSouthern"
  # [10] "SouthAfrica_Zulu"          "Namibia_Herero"            "Mozambique_Chopi"
  # [13] "Namibia_Nama"              "Angola_Sekele"             "Namibia_Wambo"
  # [16] "Namibia_Kgalagadi"         "DRC_Luozi"                 "DRC_Nienge"
  # [19] "DRC_Bandundu"              "SouthAfrica_Tswana"        "Uganda_Nkore"
  # [22] "Tanzania_Mixed"

  dups <- NEO_A_TI %>% find_duplicated_loci()
  NEO_A_TI <- NEO_A_TI %>% select_loci(-all_of(dups))

  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112_no_I_D.bk")) {
```

```
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112_no_I_D.gt")
  }
  NEO_A_TI <- gt_update_backingfile(NEO_A_TI,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112_no_I_D"
  )
  #gt_save(NEO_A_TI,
  #  file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__TI-2658_201112_no_I_D"
  #)
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110_no_I_D.gt")) {
  NEO_A_SE <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110_no_I_D.gt")
} else {
  NEO_A_SE <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110.bed", al
  nrow(NEO_A_SE) # 894
  count_loci(NEO_A_SE) # 2221421
  NEO_A_SE <- NEO_A_SE %>% select_loci(-all_of(c(which(show_loci(NEO_A_SE)$allele_ref %in% c("I", "D"))
  count_loci(NEO_A_SE)
  double_NA_NEO_A_SE <- which(is.na(show_loci(NEO_A_SE)$allele_ref) & is.na(show_loci(NEO_A_SE)$allele_
  NEO_A_SE <- NEO_A_SE %>% select_loci(-all_of(double_NA_NEO_A_SE))
  count_loci(NEO_A_SE)
  table(NEO_A_SE$population)
  dups <- NEO_A_SE %>% find_duplicated_loci()
  NEO_A_SE <- NEO_A_SE %>% select_loci(-all_of(dups))

  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110_no_I_D.bk")) {
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110_no_I_D.gt")
  }
  NEO_A_SE <- gt_update_backingfile(NEO_A_SE,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110_no_I_D"
  )
  #gt_save(NEO_A_SE,
  #  file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_A__SE-2209_191110_no_I_D"
  #)
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

Merge NEO__A

```
if (file.exists("../data/intermediate/rbind_data/NEO_A_SE_TI.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_A_SE_TI.gt")
```

```
  file.remove("../data/intermediate/rbind_data/NEO_A_SE_TI.bk")
  file.remove("../data/intermediate/rbind_data/NEO_A_SE_TI.rds")
}
```

```
## [1] TRUE
```

```
NEO_A <- rbind(NEO_A_SE, NEO_A_TI,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_A_SE_TI"
)
```

```
rm(NEO_A_SE, NEO_A_TI, NEO_A, NEO_F_OE)
gc()
```

```
##           used (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells 1217424 65.1    5811272 310.4   7264090  388.0
## Vcells 4186686 32.0  125328220 956.2 164029920 1251.5
```

EGAD50000000009

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2017_no_I_D.gt"))
  NEO_C_TE_2017 <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2017_n
} else {
  NEO_C_TE_2017 <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_201
  nrow(NEO_C_TE_2017) # 3
  count_loci(NEO_C_TE_2017) # 2221421
  NEO_C_TE_2017 <- NEO_C_TE_2017 %>% select_loci(-all_of(c(which(show_loci(NEO_C_TE_2017)$allele_ref %i
  count_loci(NEO_C_TE_2017)
  double_NA_NEO_C_TE_2017 <- which(is.na(show_loci(NEO_C_TE_2017)$allele_ref) & is.na(show_loci(NEO_C_T
  NEO_C_TE_2017 <- NEO_C_TE_2017 %>% select_loci(-all_of(double_NA_NEO_C_TE_2017))
  count_loci(NEO_C_TE_2017)
  table(NEO_C_TE_2017$population) # DRC_Ding DRC_Lwer DRC_Wonk

  dups <- NEO_C_TE_2017 %>% find_duplicated_loci()
  NEO_C_TE_2017 <- NEO_C_TE_2017 %>% select_loci(-all_of(dups))

  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2017_no_I_D.bk"
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2017_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2017_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2017_no_I_D.gt")
  }
  NEO_C_TE_2017 <- gt_update_backingfile(NEO_C_TE_2017,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2017_no_I_D"
  )
  #gt_save(NEO_C_TE_2017,
  #  file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2017_no_I_D"
  #)
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2019_no_I_D.gt"))
  NEO_C_TE_2019 <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2019_n
} else {
  NEO_C_TE_2019 <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_201
  nrow(NEO_C_TE_2019) # 297
  count_loci(NEO_C_TE_2019) # 2225121
  NEO_C_TE_2019 <- NEO_C_TE_2019 %>% select_loci(-all_of(c(which(show_loci(NEO_C_TE_2019)$allele_ref %i
  count_loci(NEO_C_TE_2019)
  double_NA_NEO_C_TE_2019 <- which(is.na(show_loci(NEO_C_TE_2019)$allele_ref) & is.na(show_loci(NEO_C_T
  NEO_C_TE_2019 <- NEO_C_TE_2019 %>% select_loci(-all_of(double_NA_NEO_C_TE_2019))
  count_loci(NEO_C_TE_2019)
  table(NEO_C_TE_2019$population)
  dups <- NEO_C_TE_2019 %>% find_duplicated_loci()
  NEO_C_TE_2019 <- NEO_C_TE_2019 %>% select_loci(-all_of(dups))

  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2019_no_I_D.bk"
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2019_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2019_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2019_no_I_D.gt")
  }
  NEO_C_TE_2019 <- gt_update_backingfile(NEO_C_TE_2019,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2019_no_I_D"
  )
  #gt_save(NEO_C_TE_2019,
  #  file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_C__TE-2567_201023_2019_no_I_D"
  #)
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

Merge NEO_C

```
if (file.exists("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.gt")
  file.remove("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.bk")
  file.remove("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.rds")
}
```

```
## [1] TRUE
```

```
NEO_C <- rbind(NEO_C_TE_2017, NEO_C_TE_2019,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_C_TE_2017_2019"
)
```

```
rm(NEO_C_TE_2017, NEO_C_TE_2019, NEO_C)
gc()
```

```
##           used (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells 1217441 65.1    4649019 248.3   7264090  388.0
## Vcells 4187294 32.0  120379092 918.5 164029920 1251.5
```

EGAD50000000010

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308_no_I_D.gt")) {
  NEO_D_RK <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308_no_I_D.gt")
} else {
  NEO_D_RK <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308.bed",
    quiet = TRUE,
    allow_duplicates = TRUE,
    valid_alleles = c("T", "G", "A", "C", "I", "D")
  )
  nrow(NEO_D_RK) # 46
  count_loci(NEO_D_RK) # 2221421
  NEO_D_RK <- NEO_D_RK %>% select_loci(-all_of(c(which(show_loci(NEO_D_RK)$allele_ref %in% c("I", "D"))
  count_loci(NEO_D_RK)
  double_NA_NEO_D_RK <- which(is.na(show_loci(NEO_D_RK)$allele_ref) & is.na(show_loci(NEO_D_RK)$allele_a
  NEO_D_RK <- NEO_D_RK %>% select_loci(-all_of(double_NA_NEO_D_RK))
  count_loci(NEO_D_RK)
  table(NEO_D_RK$population) # DRC_BanguBangu       DRC_Luba       DRC_Rega        DRC_Shi
  dups <- NEO_D_RK %>% find_duplicated_loci()
  NEO_D_RK <- NEO_D_RK %>% select_loci(-all_of(dups))

  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308_no_I_D.bk")) {
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308_no_I_D.gt")
  }
  NEO_D_RK <- gt_update_backingfile(NEO_D_RK,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308_no_I_D"
  )
  #gt_save(NEO_D_RK,
  #  file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__RK-2011_190308_no_I_D"
  #)
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

```
if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110_no_I_D.gt")) {
  NEO_D_SE <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110_no_I_D.gt")
} else {
  NEO_D_SE <- gen_tibble("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110.bed", qu
  nrow(NEO_D_SE) # 105
  count_loci(NEO_D_SE) # 2221421
```

```
  NEO_D_SE <- NEO_D_SE %>% select_loci(-all_of(c(which(show_loci(NEO_D_SE)$allele_ref %in% c("I", "D"))
  count_loci(NEO_D_SE)
  double_NA_NEO_D_SE <- which(is.na(show_loci(NEO_D_SE)$allele_ref) & is.na(show_loci(NEO_D_SE)$allele_a
  NEO_D_SE <- NEO_D_SE %>% select_loci(-all_of(double_NA_NEO_D_SE))
  count_loci(NEO_D_SE)
  table(NEO_D_SE$population) # DRC_Shi
  dups <- NEO_D_SE %>% find_duplicated_loci()
  NEO_D_SE <- NEO_D_SE %>% select_loci(-all_of(dups))

  if (file.exists("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110_no_I_D.bk")) {
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110_no_I_D.bk")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110_no_I_D.rds")
    file.remove("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110_no_I_D.gt")
  }
  NEO_D_SE <- gt_update_backingfile(NEO_D_SE,
    backingfile = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110_no_I_D"
  )
  #gt_save(NEO_D_SE,
  #  file_name = "../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_D__SE-2209_191110_no_I_D"
  #)
}
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

Merge NEO_D

```
if (file.exists("../data/intermediate/rbind_data/NEO_D_RK_SE.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_D_RK_SE.gt")
  file.remove("../data/intermediate/rbind_data/NEO_D_RK_SE.bk")
  file.remove("../data/intermediate/rbind_data/NEO_D_RK_SE.rds")
}
```

```
## [1] TRUE
```

```
NEO_D <- rbind(NEO_D_RK, NEO_D_SE,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_D_RK_SE"
)
```

```
rm(NEO_D_RK, NEO_D_SE, NEO_D)
gc()
```

```
##            used (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells  1217465 65.1    4495058 240.1   7264090  388.0
## Vcells  4187871 32.0  115767552 883.3 164029920 1251.5
```

Merge Neo datasets by set

```
rm(list = ls(pattern = "^double_NA_NEO"), dups)
```

```
## Warning in rm(list = ls(pattern = "^double_NA_NEO"), dups): object 'dups' not
## found
```

[63719:63719:20250902,122516.122931:ERROR process_memory_range.cc:86] read out of range

Merge Neo_A and Neo_B

```
if (file.exists("../data/intermediate/rbind_data/NEO_AB.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_AB.gt")
  file.remove("../data/intermediate/rbind_data/NEO_AB.bk")
  file.remove("../data/intermediate/rbind_data/NEO_AB.rds")
}
```

```
## [1] TRUE
```

```
NEO_A <- gt_load("../data/intermediate/rbind_data/NEO_A_SE_TI.gt")
NEO_B <- gt_load("../data/intermediate/rbind_data/NEO_B_SE_TE.gt")

NEO_AB <- rbind(NEO_A, NEO_B,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_AB"
)
```

```
rm(NEO_A, NEO_B, NEO_AB)
gc()
```

```
##           used (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells 1217716 65.1    4347256 232.2   7264090  388.0
## Vcells 4188491 32.0  111200850 848.4 164029920 1251.5
```

Remove intermediate merged files NEO_A and NEO_B

```
if(file.exists("../data/intermediate/rbind_data/NEO_A_SE_TI.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_A_SE_TI.gt")
  file.remove("../data/intermediate/rbind_data/NEO_A_SE_TI.bk")
  file.remove("../data/intermediate/rbind_data/NEO_A_SE_TI.rds")
}
```

```
## [1] TRUE
```

```
if(file.exists("../data/intermediate/rbind_data/NEO_B_SE_TE.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_B_SE_TE.gt")
  file.remove("../data/intermediate/rbind_data/NEO_B_SE_TE.bk")
  file.remove("../data/intermediate/rbind_data/NEO_B_SE_TE.rds")
}
```

```
## [1] TRUE
```

Merge Neo_C and Neo_D

```
if (file.exists("../data/intermediate/rbind_data/NEO_CD.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_CD.gt")
  file.remove("../data/intermediate/rbind_data/NEO_CD.bk")
  file.remove("../data/intermediate/rbind_data/NEO_CD.rds")
}
```

```
## [1] TRUE
```

```
NEO_C <- gt_load("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.gt")
NEO_D <- gt_load("../data/intermediate/rbind_data/NEO_D_RK_SE.gt")

NEO_CD <- rbind(NEO_C, NEO_D,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_CD"
)
```

```
rm(NEO_C, NEO_D)
gc()
```

```
##            used  (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells  3329824 177.9    5256710 280.8   7264090  388.0
## Vcells 22168609 169.2  106816816 815.0 164029920 1251.5
```

Remove intermediate merged files NEO_C and NEO_D

```
if(file.exists("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.gt")
  file.remove("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.bk")
  file.remove("../data/intermediate/rbind_data/NEO_C_TE_2017_2019.rds")
}
```

```
## [1] TRUE
```

```
if(file.exists("../data/intermediate/rbind_data/NEO_D_RK_SE.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_D_RK_SE.gt")
  file.remove("../data/intermediate/rbind_data/NEO_D_RK_SE.bk")
  file.remove("../data/intermediate/rbind_data/NEO_D_RK_SE.rds")
}
```

```
## [1] TRUE
```

Merge Neo_AB and Neo_CD

```
if (file.exists("../data/intermediate/rbind_data/NEO_ABCD.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_ABCD.gt")
  file.remove("../data/intermediate/rbind_data/NEO_ABCD.bk")
  file.remove("../data/intermediate/rbind_data/NEO_ABCD.rds")
}
```

```
## [1] TRUE
```

```
NEO_AB <- gt_load("../data/intermediate/rbind_data/NEO_AB.gt")

NEO_ABCD <- rbind(NEO_AB, NEO_CD,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_ABCD"
)
```

```
rm(NEO_AB, NEO_CD)
gc()
```

```
##            used  (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells  3137334 167.6    5256711 280.8   7264090  388.0
## Vcells 20405112 155.7  102776759 784.2 164029920 1251.5
```

Remove intermediate merged files NEO_AB and NEO_CD

```
if(file.exists("../data/intermediate/rbind_data/NEO_AB.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_AB.gt")
  file.remove("../data/intermediate/rbind_data/NEO_AB.bk")
  file.remove("../data/intermediate/rbind_data/NEO_AB.rds")
}
```

```
## [1] TRUE
```

```
if(file.exists("../data/intermediate/rbind_data/NEO_CD.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_CD.gt")
  file.remove("../data/intermediate/rbind_data/NEO_CD.bk")
  file.remove("../data/intermediate/rbind_data/NEO_CD.rds")
}
```

```
## [1] TRUE
```

Merge Neo_ABCD and NEO_E

```
if (file.exists("../data/intermediate/rbind_data/NEO_ABCDE.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_ABCDE.gt")
  file.remove("../data/intermediate/rbind_data/NEO_ABCDE.bk")
  file.remove("../data/intermediate/rbind_data/NEO_ABCDE.rds")
}
```

```
## [1] TRUE
```

```
NEO_ABCD <- gt_load("../data/intermediate/rbind_data/NEO_ABCD.gt")
NEO_E <- gt_load("../data/intermediate/rbind_data/NEO_E_TC_A_B.gt")

NEO_ABCDE <- rbind(NEO_ABCD, NEO_E,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_ABCDE"
)
```

```
rm(NEO_ABCD, NEO_E)
gc()
```

```
##            used  (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells  3136519 167.6    5256711 280.8   7264090  388.0
## Vcells 20396693 155.7   98729688 753.3 164029920 1251.5
```

Remove intermediate file NEO_ABCD

```
if(file.exists("../data/intermediate/rbind_data/NEO_ABCD.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_ABCD.gt")
  file.remove("../data/intermediate/rbind_data/NEO_ABCD.bk")
  file.remove("../data/intermediate/rbind_data/NEO_ABCD.rds")
}
```

```
## [1] TRUE
```

Merge Neo_ABCDE and NEO_F

```
if (file.exists("../data/intermediate/rbind_data/NEO_full.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_full.gt")
  file.remove("../data/intermediate/rbind_data/NEO_full.bk")
  file.remove("../data/intermediate/rbind_data/NEO_full.rds")
}
```

```
## [1] TRUE
```

```
NEO_ABCDE <- gt_load("../data/intermediate/rbind_data/NEO_ABCDE.gt")
NEO_F <- gt_load("../data/raw/data_input/ega-box-1667_Raw_AfricanNeo_F__OE-0808_150625_no_I_D.gt")
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

```
NEO_full <- rbind(NEO_ABCDE, NEO_F,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = TRUE, backingfile = "../data/intermediate/rbind_data/NEO_full"
)
```

```
rm(NEO_ABCDE, NEO_F)
gc()
```

```
##            used  (Mb) gc trigger  (Mb)  max used   (Mb)
## Ncells  2620750 140.0    6779997 362.1   7264090  388.0
## Vcells 15401497 117.6   94844500 723.7 164029920 1251.5
```

Remove intermediate merged file NEO_ABCDE

```
if(file.exists("../data/intermediate/rbind_data/NEO_ABCDE.gt")) {
  file.remove("../data/intermediate/rbind_data/NEO_ABCDE.gt")
  file.remove("../data/intermediate/rbind_data/NEO_ABCDE.bk")
  file.remove("../data/intermediate/rbind_data/NEO_ABCDE.rds")
}
```

```
## [1] TRUE
```

```
count_loci(NEO_full)
```

```
## [1] 1400891
```

```
nrow(NEO_full)
```

```
## [1] 1763
```

Recording R environment

Part of ensuring our workflow is reproducible is to keep track of the R environment. We will automatically save our R environment into a file called `r_environment.json`.